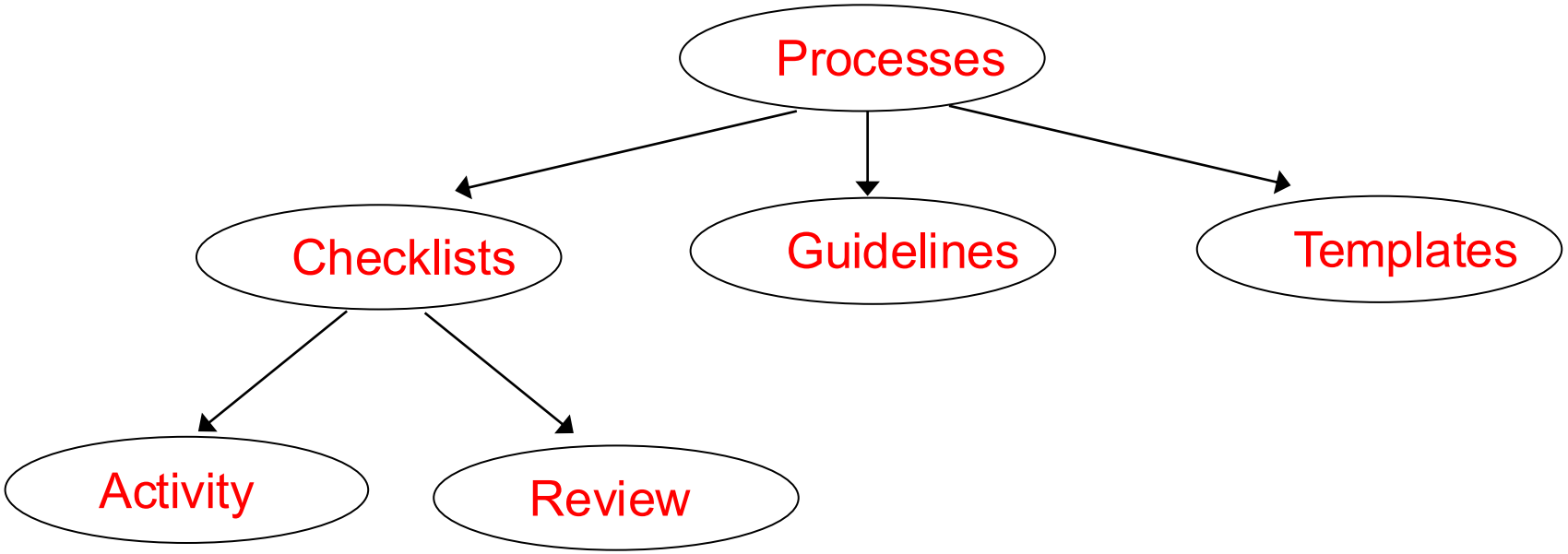
Software Project Management

Process

Process

- A process encapsulates an organization's experience in form of successful recipes.
- Process descriptions, generally, contain the sequence of steps to be executed, who executes them, the entry/exit criteria for major steps, etc.
- Guidelines, checklists, and templates provide support to use the processes.



Putting a Process in Place

- Choosing a Process.
  - All projects have a process, unfortunately some don't specify and implement their process.
  - Projects with no specified process end up thrashing.
  - Thrashing, unproductive work, can quickly damage a project.
- Generally, there are two choices for choosing a process:
  1. Tailor the organizational process to your project.
     - Used when most of the people are from the same group as before
     - Used when the last project was successful.
  2. Specify a process for your project.
     - Good when people are from different organizations using different processes

Tailoring a Process

- Steps to Tailoring an Organizational Process:
  1. Determine how your project differs from the typical organizational project.
  2. Form two lists: activities your project needs from the organizational process and tasks your project doesn't need from the process
  3. Propose changes to the organizational process
  4. Circulate the tailored process within the team and other key personnel for review and input.
  5. Integrate the changes and move quickly for closure.

Assessing the Process

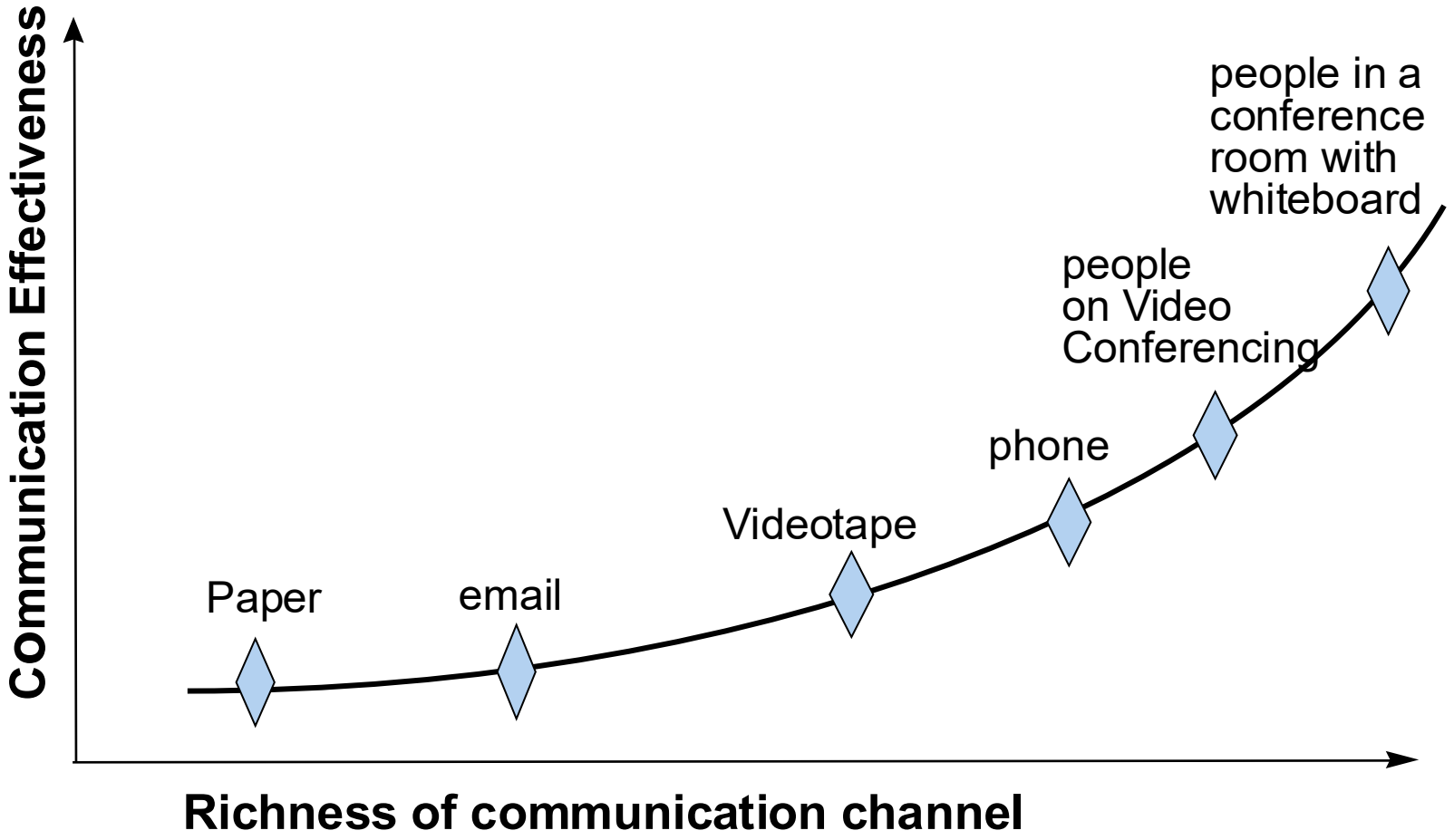
- Assessing should be an ongoing process through out the project.
- Both the project and the process should lend themselves to assessment and improvement.
- Make gathering measurements part of concurrent documentation.
- Gather data to answer the following:
  - Were the tasks and supporting activities effective?
  - How much effort did each task and activity require?
  - What tasks and activities were performed but weren't in the process specification?
  - How did the products change over time?
  - When did tasks and activities start and stop?
  - How did tasks and activities integrate?
  - When in the project did we spend effort doing what?
- Repeat this during project close out.

The Project Manager: Responsibilities

- Project planning
- Managing the project
- Lead project team
- Building client partnerships
- Targeting to the business

Few Rules Before We Embark

And finally, communicate, communicate, and communicate!



Recap

Definition of a Project

- A *project* is a sequence of unique, complex, and connected activities having one goal or purpose and that must be completed by a specific time, within budget, and according to specification.

What is a Program?

- A *program* is a collection of projects.
- The projects must be completed in a specific order for the program to be considered complete. Because they compromise multiple projects, they are larger in scope than a single project.

Project Parameters

- Five constraints operate on every project:
  - Scope
  - Quality
  - Cost
    - Time
    - Resources
- A change in one of these constraints can cause a change in another constraint to restore the equilibrium of the project
- Let's discuss each one of these in detail …

Project Parameters

Scope

- *Scope* is a statement that defines the boundaries of the project. It tells not only what will be done but also what will not be done.
- In the information systems industry, scope is often referred to as a functional specification.
- In the engineering profession, it is generally called a statement of work.

Quality

- Two types of quality are part of every project:
  - The first is ***product quality***. This refers to the quality of the deliverable form of the project.
  - The second type of quality is ***process quality***, which is the quality of the project management itself. The focus is on how well the project management process works and how can it be improved. Continuous quality improvement and process quality management are the tools used to measure process quality.

Project Parameters

Cost – The X-amount of dollars that it will cost to do the project is another variable that defines the project; the budget that has been established for the project.

- This is an important factor for projects that create deliverables that are sold to external customers

Time – The customer specifies a timeframe within which the project must be completed.

- Cost and time are inversely related to one another. The time a project takes to be completed can be reduced, but cost increases as a result.

Resources – Resources are assets, such as people, equipment, physical facilities, or inventory, that have limited availabilities, can be scheduled, or can leased from an outside party. Some are fixed, others are variable only in the long term. In any case, they are central to the scheduling of project activities and the orderly completion of the project.

Phases of the Project Management

- There are five phases of the project management life cycle:
  - **Scope/Define/Initiate** – Scope the project
  - **Plan** – Develop the project plan
  - **Execute** – Launch the plan
  - **Monitor** – Monitor/control project progress
  - **Close** – Close out the project
- Note: these can be repeated for each phase
- Each process/phase/activity is described by:
  - Inputs
  - Tools & Techniques
  - Outputs

Initiating Process

- Develop project charter
  - State the problem/opportunity.
  - Concerned with authorizing a project
  - May be used for a whole project
  - May be used for a single project phase in a large, multiphase project
- Develop preliminary project scope statement
  - Concerned with producing a preliminary, high-level definition of project
  - Broadly defines what is and what is not part of the project
- Establish the project plan.
  - Define the project objectives.
  - Identify the success criteria.
  - List assumptions, risks, obstacles

Initiating Process

- Inputs
  - Product Description
  - Strategic plan
  - Project Selection Criteria
  - Historical Information
- Outputs
  - Project Charter
  - Project Manager assigned
  - Constraints
  - Assumptions

Planning Process

Devising and maintaining a workable scheme to accomplish the business need that the project was undertaken to address

- Scope Planning
- Scope Definition
- Activity Definition
- Activity Sequencing
- Activity Duration Estimating
- Resource Planning
- Cost Estimating
- Cost Budgeting
- Schedule Development
- Quality Planning
- Communications Planning
- Organization Planning
- Staff Acquisition
- Risk Planning
- Procurement Planning
- Project Plan Development

Develop the project plan

- Develop project management plan
  - Concerned with creating and integrating all sub-plans into a single source of information
  - Identify the project activities.
- Scope planning
  - Concerned with how the project scope statement will be created
- Create WBS
- Scope definition
  - Concerned with actual creation of project scope statement
- Activity definition
  - Activity sequencing
  - Activity duration estimating
  - Activity resource estimating
- Determine resource requirements.

Planning processes

- Schedule development
  - Concerned with analyzing activity outputs (definition, etc.) to create project schedule
- Construct/analyze the project network.
- Cost estimating **
- Cost budgeting
  - Concerned with aggregating costs of individual activities to establish cost baseline
- Quality planning *
  - Concerned with quality standards and how to achieve them
- Human resource planning *
- Communications planning *

* indicates minimal or no coverage

** indicates optional coverage

Planning processes

- Risk management planning
  - Concerned with how to carry out risk management activities
  - Risk identification
  - Qualitative risk analysis
    - Concerned with prioritizing risks based on probability of occurrence and impact
  - Quantitative risk analysis *
  - Risk response planning
    - Concerned with mitigating risks to project objectives
- Plan purchases and acquisitions *
  - Concerned with what, when, and how of purchases and acquisitions
- Plan contracting *
- Prepare the project proposal.

Executing Process

Coordinating people and other resources to carry out the plan

- Project Plan Execution
- Scope Verification
- Quality Assurance
- Acquire project team
  - Identify and organize the project team.
  - Establish team operating rules.
  - Team Development
- Solicitation
  - Information Distribution
  - Source Selection
  - Contract Administration
- Level project resources.
- Schedule work packages.
- Document work packages.

Monitoring & Controlling Process

Monitor and control project work

- Ensuring that project objectives are met by monitoring and measuring progress and taking corrective measures when necessary
- Concerned with acquiring and assessing performance information to effect process improvements

- Integrated change control
  - Overall Change Control
  - Scope Change Control
  - Schedule Control
- Scope control – Concerned with changes to project scope
- Scope verification – Concerned with acceptance of project deliverables
- Schedule control – Concerned with changes to project schedule

Monitoring & Controlling Process

- Cost control * – Concerned with changes to the project budget
- Quality Control – Concerned with monitoring quality compliance of project results and correcting unsatisfactory results
- Manage project team – Concerned with tracking performance, providing feedback, and coordinating changes
  - Define problem-escalation process.
  - Monitor project progress versus plan.
  - Establish progress reporting systems.
  - Performance reporting * – Concerned with status, progress, and forecasting
- Install change control tools/process.
- Risk monitoring and control
- Manage stakeholders
- Contract administration *

☛ Revise project plans.

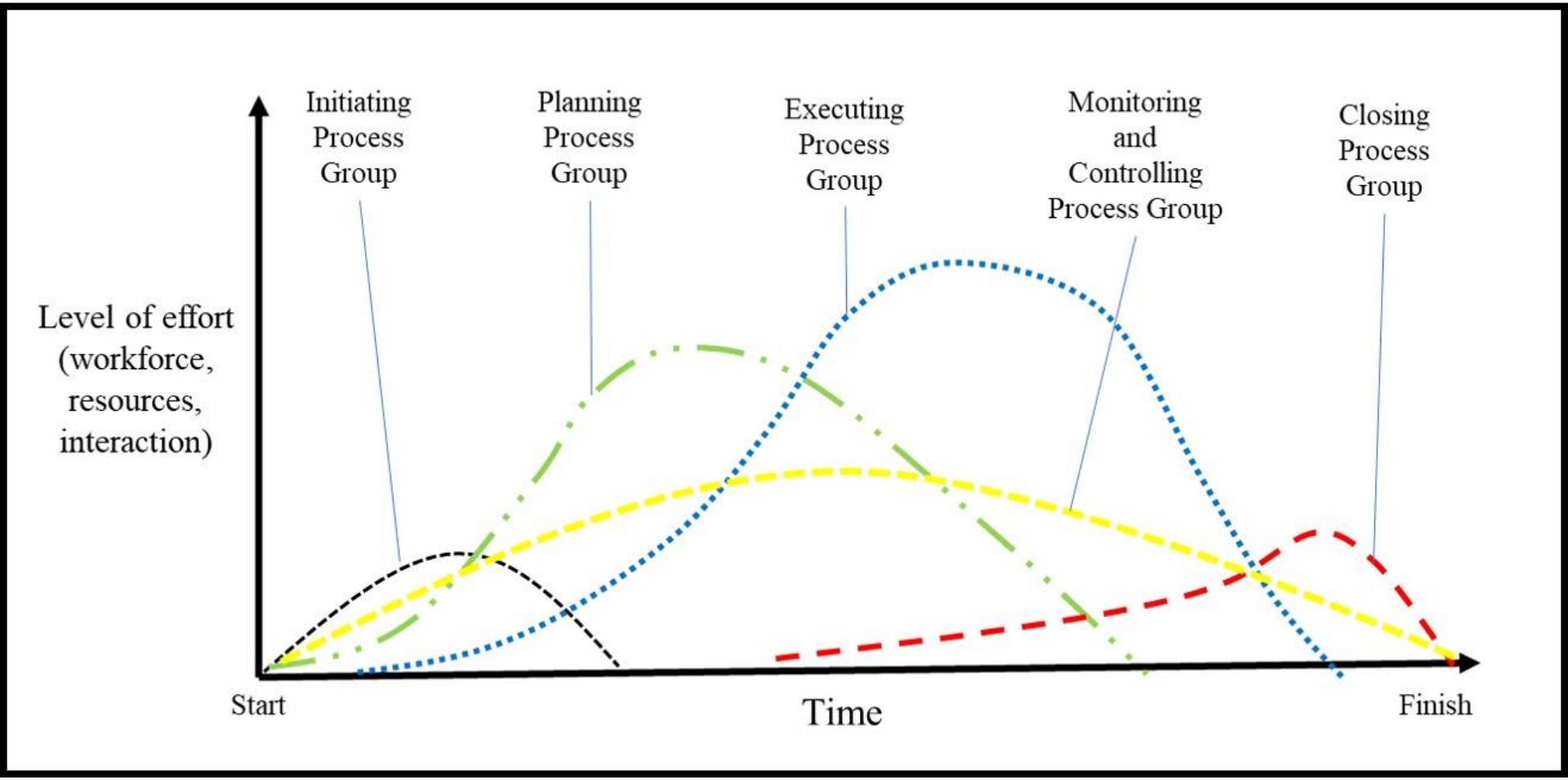
Close out the project

Formalizing acceptance of the project or phase and bringing it to an orderly end

- Administrative Closure
  - Concerned with finalizing all activities across all Process Groups
  - Complete project documentation.
  - Complete post-implementation audit.
    - Lessons learned
  - Issues final project report.
- Contract Close-out
  - Concerned with completing and settling all contracts
  - Obtain client acceptance.
  - Install project deliverables.

Phases of the Project Management

Level of Activity and Overlap of Process Groups Over Time



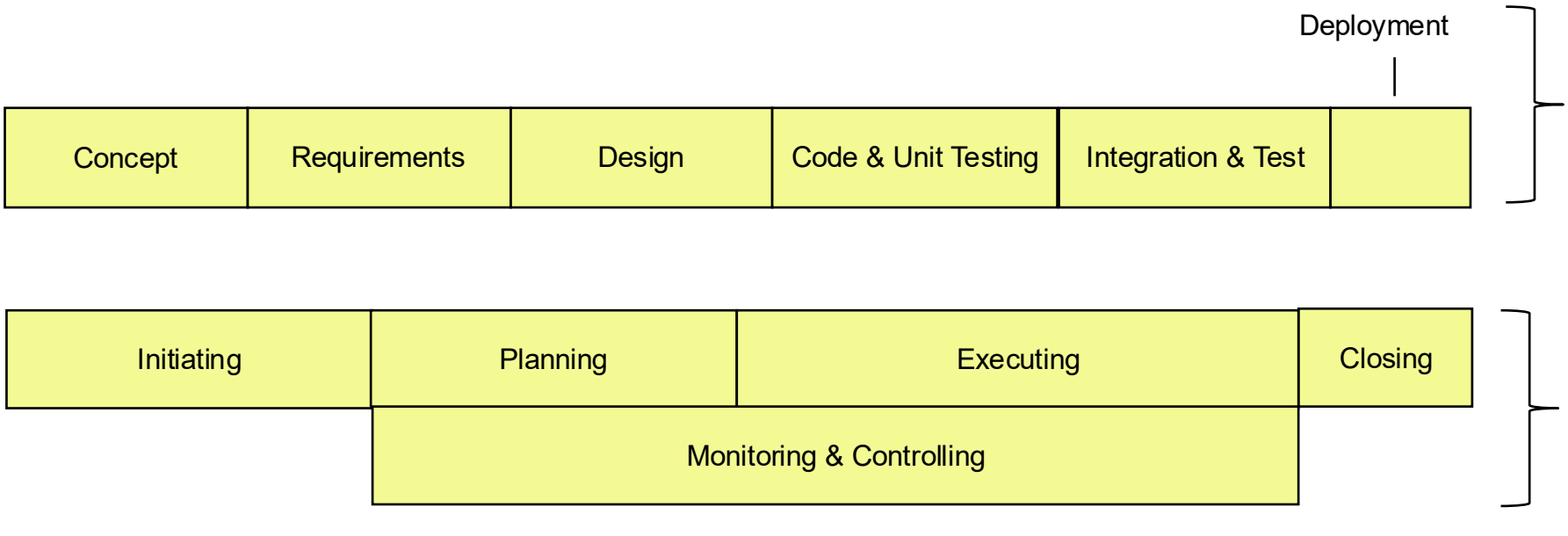
Project Processes & Their Integration

- Project Management Processes (Principles of Project Management)
  - Initiating processes (Defining)
  - Planning processes
  - Executing processes
  - Monitoring & controlling processes
  - Closing processes
- System Development Processes (Iterative/evolutionary)
  - Inception phase
  - Elaboration phase
  - Construction phase
  - Transition phase
- Integrating IT Project Processes
  - PM/IT project integration tactics

PM/IT process integration tactics

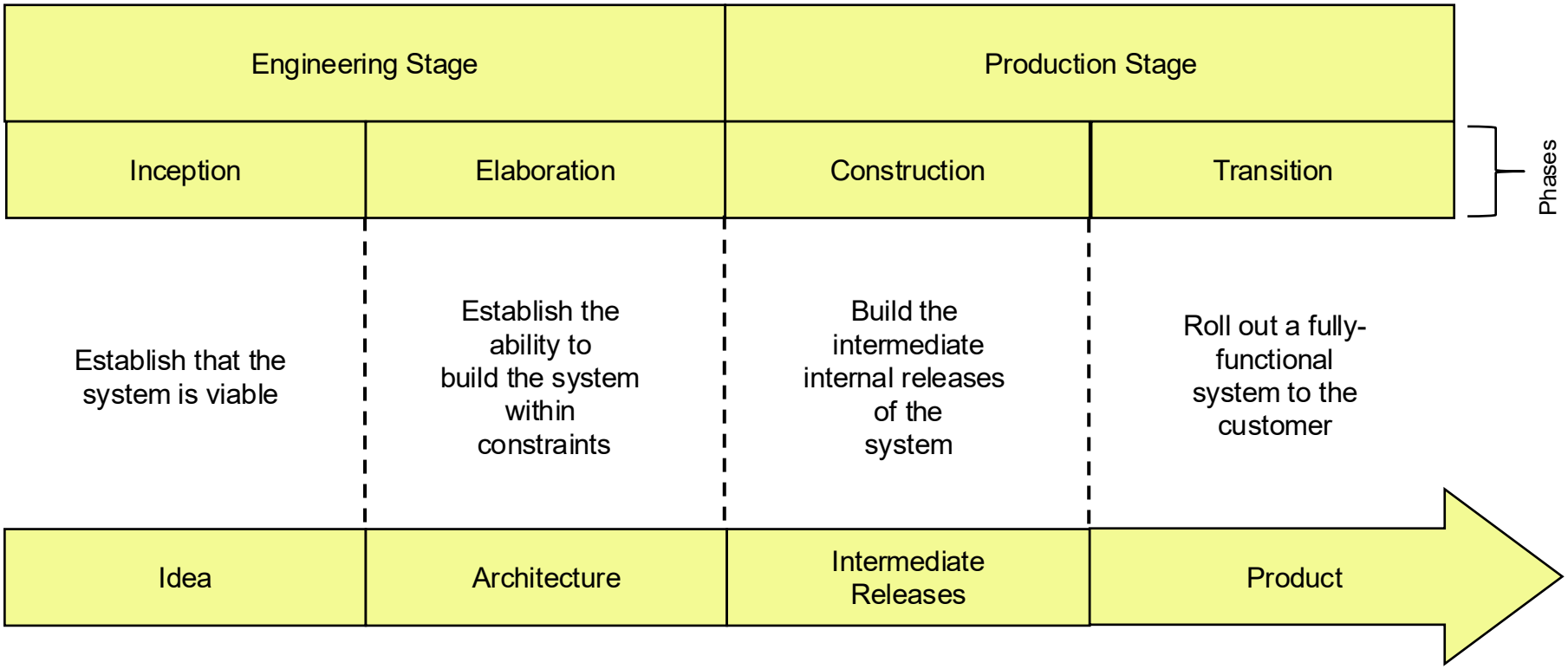
- Wherever possible, establish common policies, processes, and procedures between IT and PM groups
- Identify an integration manager to link IT and PM groups
- Use a common, integrated, consistent vocabulary that is continuously updated to facilitate inter- (as well as intra-) group communications
- Ensure that project manager possesses suitable process integration skills and is familiar with IT risks
- Involve IT analysts in development of business requirements
- Identify an ombudsman to quickly resolve issues that arise between PM and IT groups

Project & SDLC integration waterfall development model



Phases in iterative* system life cycle

The stages below are repeated (iterative) – see notes



* I often interchange *iterative* & *evolutionary*

Project & SDLC integration
iterative/incremental development model

